

Scalability & Future Plan

Date	11 July 2026
Team ID	XXXXXX
Project Name	Smart Lender - AI Powered Loan Eligibility Prediction
Maximum Marks	1 Mark

This document outlines how the Smart Lender solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing the project beyond its current state.

Current System Limitations

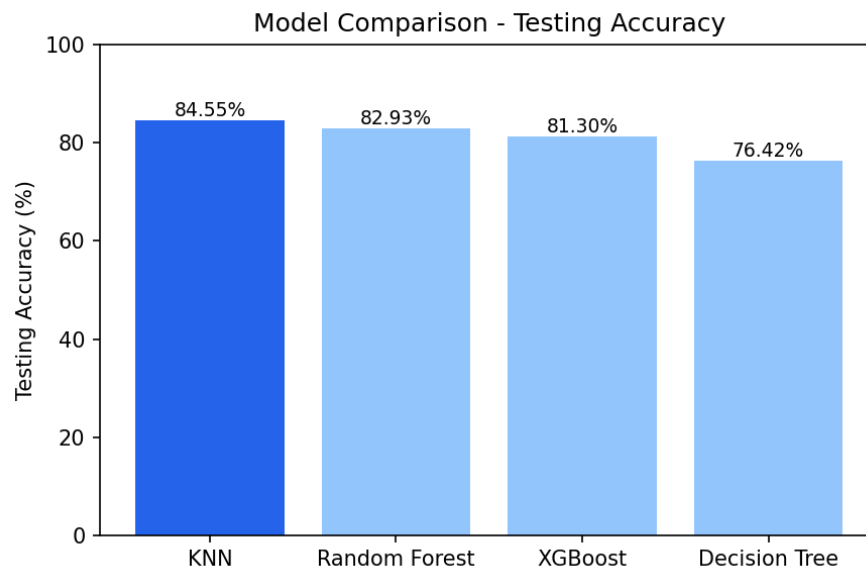
S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Trained on a static historical dataset (614 records)	Predictions may not reflect the latest lending patterns	High
2	No user authentication or role-based access	Anyone with the URL can submit predictions	Medium
3	Single Flask instance, no caching or database	Limited scalability under high concurrent load	High
4	No explainability (SHAP/LIME) for individual predictions	Applicants/officers cannot see why a decision was made	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local/Gunicorn instance	Deploy behind a load balancer with multiple Gunicorn workers / auto-scaling
2	Data Storage	Static CSV files	Migrate to a relational database (PostgreSQL/MySQL) for applicant records
3	Model Freshness	Model trained once, not retrained	Add a scheduled retraining pipeline as new loan outcomes become available
4	Security	Basic input validation only	Add authentication, HTTPS, and rate limiting on the /predict endpoint
5	Explainability	Confidence score only	Integrate SHAP/LIME to show feature-level explanations for each prediction

Model Comparison Reference

The chart below (generated from models/model_comparison.csv) shows why KNN was selected for deployment and provides a baseline for any future model upgrades.



Future Enhancements

- Explainable AI (SHAP/LIME) for transparent decisions
- Loan default probability estimation (beyond binary approval)
- User authentication and role-based dashboards for loan officers
- Database integration for applicant history and audit trails
- Cloud-native deployment with auto-scaling
- Email/SMS notification of prediction results
- Automated PDF report generation per application